Supremum & Infimum

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1 Definitions

1.1 Upward/downward closed set

Definition 1.1. (Upward closed set)

$$\uparrow_{(P,\leq)} x := \{y \in P | x \leq y\}$$

Definition 1.2. (Downward closed set)

$$\downarrow_{(P,\leq)} x := \{y \in P | y \leq x\}$$

1.2 Ordered set of majorants/minorants

Definition 1.3. (Ordered set of majorants)

$$\mathrm{Ma}_{(P,\leq)} S := \left(P \cap \bigcap \uparrow_{(P,\leq)} [S], \leq\right)$$

Definition 1.4. (Ordered set of minorants)

$$\mathrm{Mi}_{(P,\leq)} S := \left(P \cap \bigcap \downarrow_{(P,\leq)} [S], \leq\right)$$

1.3 Ordered set of greatest/least elemenents

Definition 1.5. (Ordered set of greatest elements)

$$\mathrm{Greatest}_{(P,\leq)} := \mathrm{Ma}_{(P,\leq)} P$$

Definition 1.6. (Ordered set of least elements)

$$\mathrm{Least}_{(P,\leq)} := \mathrm{Mi}_{(P,\leq)} P$$

1.4 Ordered set of supremums/infimums

Definition 1.7. (Ordered set of supremums)

$$\mathrm{Sup}_{(P,\leq)} S := \mathrm{Least}_{\mathrm{Ma}_{(P,\leq)} S}$$

Definition 1.8. (Ordered set of infimums)

$$\mathrm{Inf}_{(P,\leq)} S := \mathrm{Greatest}_{\mathrm{Mi}_{(P,\leq)} S}$$

2 some bad older definitions, but good thms

Definition 2.1. bad

$$\mathrm{UB}_R(S) = R[S]$$

Definition 2.2. bad

$$\mathrm{LB}_R(S) = R^{-1}[S]$$

Context and notation:
Here by $\leq$ we denote the standard relation "less or equal to" on real numbers.
$\leq$ for inequalities of numbers
$\leqslant$ for inequalities of sets of numbers
$R := \leq$
$\leq = \geq^{-1}$
$\geq = \leq^{-1}$
Remark: if $\leq$ is transitive then $\leqslant$ is transitive on non-empty(sic!) sets.

Definition 2.3. good

$$A \leqslant B \iff \forall a \in A \forall b \in B : a \leq b$$

Definition 2.4. bad

$$\mathrm{UB}(S) = \leq [S]$$

Definition 2.5. bad

$$\mathrm{LB}(S) = \geq [S]$$

Theorem 2.1. maybe

$$A \leqslant B \iff B \subseteq \mathrm{UB}(A)$$

Theorem 2.2. maybe

$$A \leqslant B \Longleftrightarrow A \subseteq \mathrm{LB}(B)$$

Definition 2.6. bad

$$\mathrm{Sup}(S) = \mathrm{UB}(S) \cap \mathrm{LB}(\mathrm{UB}(S))$$

Definition 2.7. bad

$$\mathrm{Inf}(S) = \mathrm{LB}(S) \cap \mathrm{UB}(\mathrm{LB}(S))$$

Definition 2.8. maybe

$$\sup(S) = (\iota x : x \in \mathrm{Sup}(S))$$

Definition 2.9. maybe

$$\inf(S) = (\iota x : x \in \mathrm{Inf}(S))$$

Theorem 2.3. good

$$A \leqslant B \Longrightarrow A \leqslant \mathrm{Sup}(A) \leqslant \mathrm{Inf}(B) \leqslant B$$

Theorem 2.4. doubt

$$(a : \mathbb{N} \to \mathbb{R}) \wedge (\{a_n\}_n \text{bdd above in } \mathbb{R}) \wedge ((a_n)_n \text{ non-decreasing}) \longrightarrow (a \longrightarrow \sup\{a_n\}_n)$$